WHICH CONDITIONING RECORDING DOES A VOICE CLONE FOLLOW? A BALANCED CROSS-MICROPHONE CROSSOVER

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ABSTRACT

Voice-clone attribution usually stops at the generator or speaker. We ask a finer question: which of two recordings of the same speaker conditioned the clone? In a pre-registered complete crossover, a fixed 54-speaker VCTK roster provides two events with paired microphone captures. Four open cloning systems and four fixed, held-out texts produce 3,456 clones. Generation uses one microphone; attribution compares the two events through the other, so speaker identity, generated text, exact waveform and a shared microphone fingerprint cannot decide the label. With fixed speaker-verification readouts, the mic1→mic2 follow rate is .896 [.869,.921] for ECAPA and .622 [.593,.652] for WavLM; the predeclared reverse direction gives .907 [.881,.931] and .620 [.589,.653]. The ECAPA result meets the pre-registered large-effect criterion; WavLM corroborates its direction, which persists when the microphone roles are reversed. This establishes closed-set attribution between these two fixed pickup paths, not the carrier mechanism or open-set presence detection.

***Index Terms*—** voice cloning, source attribution, audio provenance, speaker verification

1. INTRODUCTION

Zero-shot speech generation conditions on a short reference recording. When that recording was used without consent, identifying the generator or cloned speaker does not answer the narrower provenance question: *which recording was used?* Existing source tracing attributes a synthesis pipeline [1, 2], and clone attribution identifies an enrolled speaker [3, 4, 5]. Recording-level evidence would instead connect an output to one conditioning event among other recordings of the same voice.

That target is easy to confound. A clone and candidate share speaker identity; one candidate may be intrinsically closer to every clone; recording session, duration or microphone can reveal the answer; and generated content may overlap. Prior prompt-transfer work shows preservation of prosody and acoustic environment [6, 7, 8], while speaker embeddings themselves encode session and channel [9, 10]. High retrieval from a same-speaker pool therefore does not by itself show that the target follows the recording used for conditioning.

We turn the question into a balanced intervention. For each speaker, clone from recording event A and from event B, and test both signed comparisons. Prompt and candidate are paired captures from different microphones. Four systems and texts are fully crossed; ECAPA-TDNN [11] is fixed as primary and a speaker-verification-fine-tuned WavLM [12] as cross-encoder corroboration. Within a 54-speaker complement inherited from ECAPA-informed development rosters, all speakers are retained and pair selection is frozen from metadata alone, without clone outcomes.

Our contributions are: (1) a complete A/B crossover that identifies whether a clone follows its conditioning recording while changing pickup path; (2) a pre-registered, independently audited 3,456-clone evaluation on 54 speakers, four systems and two fixed representations; and (3) a boundary result: recording-event attribution is consistent in a two-candidate closed set, but neither the physical carrier nor deployable recording-presence detection follows.

2. RELATED WORK

Attribution and provenance. Generator tracing, shared-generator verification and speaker attribution operate above the individual recording [1, 2, 3, 4]. Watermarking requires instrumentation before misuse [13]; membership inference asks whether audio entered training [14], not whether it was the inference-time prompt. To our knowledge, no prior controlled study intervenes on the prompt recording within speaker and then replaces the candidate pickup path.

Prompt transfer and nuisance information. Prompt-conditioned systems preserve prosody and environment [6, 7, 15]; prompt cosine is also a standard synthesis metric [16]; exact prosody cloning transfers reference prosody [17]. Source-voiceprint tracing recovers identity from converted audio [18], while rank/local-disclosure work treats linkage without global thresholds [19, 20]. None intervenes on which of two same-speaker recording events conditions a clone. Concurrent work separates prompt environment from identity [21]. These works motivate recording-level leakage but do not identify the chosen event: we target *A versus B*, not generic clone–prompt resemblance.

3. CROSS-MICROPHONE CROSSOVER

3.1. Task and design

For speaker s, let recording events A and B each have paired VCTK mic1/mic2 captures [22] — a DPA 4035 omni and a Sennheiser MKH 800. That the two are genuinely distinct signals is the design's load-bearing assumption, so we tested it rather than assume it. Raw correlation does not settle it in either direction: a delayed copy can score near zero, while genuine pairs reach .92 median once delay-aligned. We therefore use the normalized least-squares residual after optimal scalar gain and a ± 50 ms zero-padded delay; its .00001 duplicate bar is a numerical tolerance for these transforms, not a universal perceptual threshold. Across all 108 event captures the residual

Primary: opposite-pickup attribution
$A_{m1} \to c_{A,m1} \to \{A_{m2}, B_{m2}\}$ and
$B_{m1} \to c_{B,m1} \to \{A_{m2}, B_{m2}\}$
Correct labels are reversed across arms; speaker, system and generated text are held fixed. The direction-reversal analysis swaps $m1$ and $m2$.

Fig. 1. Complete two-arm crossover. Generation never consumes the microphone used for the attribution candidates.

Table 1. What the crossover blocks, and the residual scientific boundary.

Shortcut	Design response	Residual
Speaker identity	A/B within speaker	none for the label
Candidate preference	both signed arms	arm interactions
Generated content	four texts shared A/B	text fixed, not sampled
Exact waveform/device	opposite microphone	event acoustics
Duration/text length	metadata-matched A/B	other local cues

is .38 median (range .21–.70). Exact, gain-scaled and delayed injections on every source, including both delay-window boundaries, are at most .000000026; no pair is byte-identical. A clone $c_{A,m1}$ is conditioned on event A's mic1 capture. The primary comparison asks whether a fixed encoder f gives

$$I[\cos(f(c_{A,m1}), f(A_{m2})) > \cos(f(c_{A,m1}), f(B_{m2}))] , \quad (1)$$

with ties worth $1/2$. We also clone from B and reverse the correct label. Thus an encoder's unconditional preference for A cannot create a positive effect. The predeclared direction reversal swaps microphones (mic2 prompt, mic1 candidates) and cannot rescue a failed primary direction.

With $d_X = s(c_X, A) - s(c_X, B)$ on opposite-microphone candidates, a cell contributes $\{I[d_A > 0] + I[d_B < 0]\}/2$. Clone-independent candidate bias therefore yields chance; ties receive $1/2$.

The threat model is deliberately narrow: the analyst has a clone and two known, same-speaker candidate recordings, exactly one of which supplied the prompt. The objective is source attribution, not deciding whether an arbitrary recording was used. Encoder parameters, score direction and the tie rule are fixed before evaluation; the analyst does not train on these speakers or systems.

3.2. Frozen roster and generation

VCTK v0.92 documents DPA 4035 and Sennheiser MKH 800 microphones in a fixed hemi-anechoic setup [22]. Its public description does not establish acquisition timing, so our claim requires paired captures, not simultaneity. The frozen manifest independently supports the pairing: all 108 selected event/microphone pairs have identical frame counts and sampling rates.

The 54-speaker roster is the exact complement of two earlier development tiers. Tier 1 applied an ECAPA-geometry gate to 105 speakers and retained the lexicographically first 30 passing speakers; Tier 2 used ECAPA within-speaker geometry and retained all 24 remaining qualifiers. Consequently, the present roster is disjoint from development and blind to EXP-205 outcomes, but not embedding-naive or population-sampled. Within that fixed complement, all 54 speakers remain and the A/B-pair rule below never inspects an embedding, score or clone outcome.

For each speaker, selection enumerates 4–10 s paired utterances, excludes a previously used seed, and requires the A/B duration gap to be at most .25 s and the relative seconds-per-UTF-8-byte duration gap at most 5% on *both* microphones. A deterministic metadata-only rule chooses the closest pair; no embedding, score or clone outcome enters within-roster pair selection. Transcripts A and B differ, but generation uses four fixed texts disjoint from both candidates and fully crossed across arms. The selected roster is substantially tighter than the eligibility limits: its worst duration gap is .112 s and worst relative duration-per-byte gap is 2.57% across either microphone.

Table 2. Speaker-weighted follow rate [95% composition-stability interval]. Same-mic columns are diagnostics, never verdict inputs.

Prompt→candidate	Encoder	Cross-mic	Same-mic diagnostic
mic1→mic2	ECAPA	**.896 [.869,.921]**	.938 [.920,.955]
	WavLM	**.622 [.593,.652]**	.631 [.602,.660]
mic2→mic1	ECAPA	**.907 [.881,.931]**	.930 [.910,.948]
	WavLM	**.620 [.589,.653]**	.659 [.627,.692]

We evaluate F5-TTS 1.1.22 [23], XTTS-v2 0.25.3 [24], CosyVoice 2 [25] and Seed-VC [26]. Seed-VC converts one fixed held-out source utterance per generation text, shared across targets. The full design is 54 speakers $\times 2$ events $\times 2$ prompt microphones $\times 4$ texts $\times 4$ systems $= 3{,}456$ clones. Exact checkpoints, repositories, environments, files and per-clone ancestry ledgers are hash-pinned.

3.3. Estimand, bars and audit

Each speaker contributes 32 dependent comparisons per direction (two arms, four systems, four texts). We first average within speaker, then give all 54 speakers equal weight. Percentile intervals resample whole speaker vectors 100,000 times with a frozen seed. They quantify stability to speaker composition in this fixed eligible roster, not population-coverage confidence. Systems and texts are fixed crossed factors; system-specific points are descriptive and receive no individual inferential claims.

The frozen primary conjunction is: **E**, ECAPA lower endpoint $> .50$; **O**, ECAPA point $\geq .80$ and lower endpoint $> .70$; and **R**, WavLM lower endpoint $> .50$. "O" is the protocol's internal label; it is an author-declared large-effect criterion, not a deployment-utility claim. The direction-reversal headline is permitted only if both reverse ECAPA and WavLM lower endpoints exceed .50. Diagnostics cannot rescue any conjunct.

Before outcomes, the manifest, analyzer, verdict and config were frozen and audited. A transport-only one-field repair was recorded and independently approved before the result was opened. A second read-only audit rehashed 3,456 clones and ledgers plus 216 candidates and independently reproduced every result field to 2×10^{-15}. The released artifact contains sanitized scores and an independent verifier that reconstructs the 3,456-row identity grid, bootstrap and verdict without audio or model weights; its execution environment is pinned.

4. RESULTS

4.1. The conditioning event survives pickup change

Table 2 reports the predeclared directions. ECAPA attributes the conditioning event in 89.6% of primary comparisons and 90.7% in reverse, meeting the .80/.70 primary bar. WavLM is much weaker but directionally consistent at 62.2%/62.0%. E, O, R and both reverse components pass: the ECAPA effect meets the frozen large-effect rule and WavLM corroborates its direction.

Same-microphone diagnostics are only modestly higher: exact pickup identity is unnecessary, though paired captures of one event do not establish microphone invariance.

Table 3. Frozen bars and observed values. LCB is the lower endpoint in Table 2; inequalities are strict except the .80 point bar.

Axis	Frozen criterion	Observed	Status
E	primary ECAPA LCB $>$ .50	.869	PASS
O	ECAPA point $\geq$ .80, LCB $>$ .70	.896/.869	PASS
R	primary WavLM LCB $>$ .50	.593	PASS
D_E	reverse ECAPA LCB $>$ .50	.881	PASS
D_R	reverse WavLM LCB $>$ .50	.589	PASS

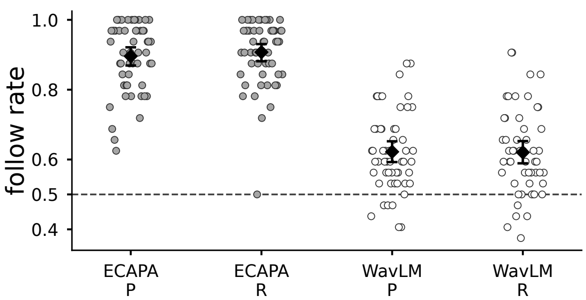

Fig. 2. Fixed-roster speaker means (54 dots/group; 32 comparisons per dot). Diamonds and bars are pooled points and 95% composition-stability intervals; the dashed line is chance. P/R denote primary/reverse directions.

Figure 2 exposes the heterogeneity hidden by pooled points. All 54 primary ECAPA speaker means exceed .50; in reverse, 53 exceed it and one equals it. WavLM is not uniformly positive—47 primary and 43 reverse speaker means exceed .50—so its contribution is corroboration of the aggregate direction, not a per-speaker guarantee. Whole-speaker resampling preserves this heterogeneity rather than treating 1,728 comparisons per direction as independent.

4.2. Breadth across fixed systems

Table 4 gives descriptive system points: ECAPA exceeds .82 throughout; WavLM spans .572–.671, with weaker XTTS-v2 cells visible. Fixed systems and no simultaneous per-system intervals license breadth only across this tested design.

The ECAPA–WavLM gap rules out near-ceiling magnitude for every representation; WavLM's positive lower endpoints still oppose ECAPA-weight specificity. Both are speaker-verification embeddings, so a shared cue remains possible. Positive continuous margins are diagnostic only.

4.3. Post-review sensitivity checks

Table 5 stratifies the inherited ECAPA-informed complement by the earlier Tier 1 gate. Direction persists among 36 pass-but-not-retained and 15 fail speakers; three absent from that record remain separate. This post-hoc check cannot create an embedding-naive sample or alter the pooled verdict.

Arm-separated points also oppose a one-candidate shortcut. For A/B, ECAPA is .889/.903 in primary and .892/.922 in reverse; WavLM is .588/.656 and .660/.581, respectively. Thus WavLM's arm asymmetry reverses with microphone direction. All 32 system$\times$direction$\times$encoder$\times$arm cells are above .50 (minimum .523), descriptively; no cell has a separate interval or multiplicity-adjusted test.

Table 4. Descriptive follow-rate points by system. P/R: primary/reverse.

Encoder, dir.	F5-TTS	XTTS-v2	CosyVoice 2	Seed-VC
ECAPA, P	.942	.833	.910	.898
ECAPA, R	.942	.822	.942	.924
WavLM, P	.618	.572	.671	.627
WavLM, R	.657	.574	.657	.593

Table 5. Post-hoc points by prior Tier 1 gate status. P/R: microphone direction; no stratum-specific confirmatory claim.

Prior gate status	n	ECAPA P/R	WavLM P/R
Pass, not retained	36	.908/.907	.609/.596
Fail	15	.879/.915	.654/.685
Not evaluated	3	.833/.875	.625/.583

4.4. What changed relative to earlier evidence

Earlier same-speaker mixed-chapter pools retrieved the seed at 9.7–16.4$\times$ chance, and same-chapter fused AUC was .815–.872. Those designs established a floor but left seed and speaker/session structure entangled; F5-TTS also exposed a duration-per-text-byte cue. A subsequent two-seed crossover obtained .806 [.757,.854] but missed its pre-registered .90 point bar and used favorable seed pairs on one pickup path. We retain that as positive discovery, not confirmation.

The present experiment is a separate test: the 54-speaker development complement, all speakers retained, within-roster metadata-only A/B pairs, balanced labels, opposite-microphone candidates and fixed readouts. Its .80/.70 ECAPA bar was frozen for this design; passing it does not retroactively change the earlier .90 miss. This trajectory matters: each control removed a shortcut rather than merely adding another positive cell.

5. BOUNDARY AND LIMITATIONS

Attribution is not presence detection. The crossover is a closed-set two-alternative question known to contain the source. In our earlier paired present/absent evaluation, normalized minDCF at a .01 prior was .79–1.00 against 1.00 for reject-all; only two of four systems had intervals excluding reject-all. A score can order A above B while lacking a stable global threshold. We therefore limit investigative triage to prioritizing candidates inside a known-positive two-recording set for human provenance review; it cannot decide whether an arbitrary queried recording was present and is not an automated accusation.

The carrier is unidentified. Opposite microphones exclude an exact waveform and shared device fingerprint, while A/B balancing excludes a fixed candidate preference. Metadata matching sharply limits raw duration and text-normalized-duration shortcuts. It does not separate event-level prosody, articulation, background acoustics, paired-capture room state or their interactions with model conditioning. A mechanism claim requires interventions on those factors.

Scope. The result concerns four fixed checkpoints, four fixed texts, read English speech and one inherited 54-speaker VCTK complement. It is disjoint from the earlier development rosters but ECAPA-informed through their composition, not globally unseen or population-sampled. Checkpoint training corpora are not documented at recording level, so VCTK exposure cannot be excluded. The intervals do not sample systems, texts or a speaker population. Natural speech, noisy field recordings, other pickup paths, more than two candidates and open-set search remain untested.

Release and misuse. Code, score-level data, provenance metadata and aggregate/per-speaker results are at `https://github.com/rvirgilli/voice-clone-cross-microphone-crossover/tree/f2-icassp2027`; model weights, source audio and playable impersonations are not redistributed. Recording attribution can assist provenance audits and also deanonymize speakers; licensing alone does not settle consent.

6. CONCLUSION

In this fixed roster, a clone follows which same-speaker recording conditioned it when prompt and candidates use the two tested microphone paths. The effect meets the pre-registered large-effect criterion with ECAPA, is smaller but corroborated by WavLM, and persists when the microphone roles are reversed, while pooling equally over four fixed systems. This is a large closed-set signal between two tested paths; it neither identifies the carrier nor solves open-set recording-presence detection.

7. REFERENCES

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